

1.4 States of Matter

Question Paper

Course	CIEA Level Chemistry
Section	1. Physical Chemistry
Topic	1.4 States of Matter
Difficulty	Easy

Time allowed: 60

Score: /46

Percentage: /100

Question 1a

This question is about gases.

i)

Describe the arrangement and movement of particles in a gas.

[3]

ii)

Explain why gases exert a pressure.

[1]

[4 marks]

Question 1b

As the volume of a gas is changed, the pressure of the gas changes. A graph of the volume of gas against $\frac{1}{\text{pressure}}$ at constant temperature is shown in Fig. 1.1.

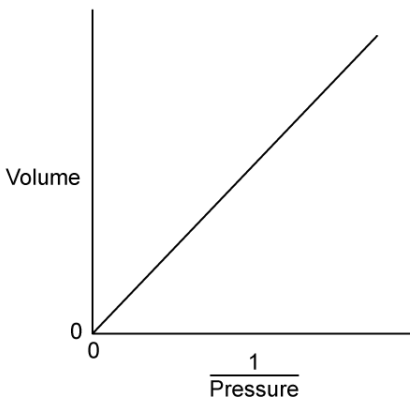


Fig. 1.1

What relationship does the graph in Fig. 1.1 show between the volume of gas and pressure at constant temperature?

[1 mark]

Question 1c

Changing the temperature of a gas, at a constant volume, will affect the volume of the gas.

Draw the relationship between the temperature and the volume of the gas on Fig 1.2.

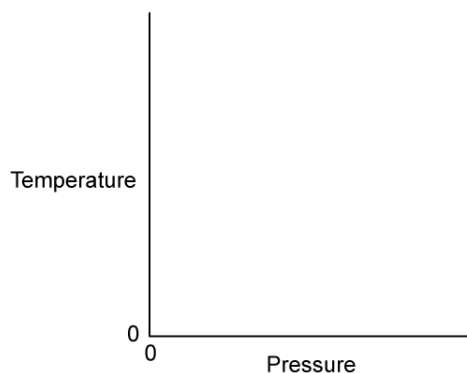


Fig. 1.2

[2 marks]

Question 2a

Two assumptions made about ideal gas behaviour are that ideal gases have zero particle volume and no intermolecular forces of attraction

Explain why helium comes very close to ideal gas behaviour.

[2 marks]

Question 2b

The ideal gas equation is $pV = nRT$ where R is the molar gas constant with a value of $8.31 \text{ J K}^{-1} \text{ mol}^{-1}$.

Explain as fully as you can the meaning of the following terms, and give the units for each to correspond with the value of R .

p

V

T

[6 marks]**Question 2c**

Calculate the volume, in dm^3 , of one mole of helium gas at 10°C and 92.0 kPa .

Give your answer to 3 significant figures.

Show your working.

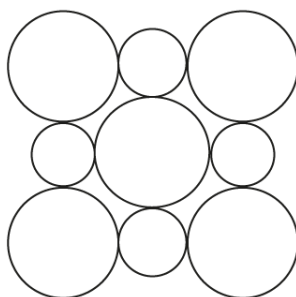
volume of one mole of helium gas = dm^3

[3 marks]

Question 3a

Sodium ions and chloride ions form a giant ionic lattice in solid form. Sodium ions are smaller than chloride ions.

Complete the diagram to show the location of the ions in the lattice of solid sodium chloride.



[2 marks]

Question 3b

Explain the strength of the ionic bond in sodium chloride.

[2 marks]

Question 3c

Magnesium fluoride is another compound that forms a giant ionic lattice in solid form.

Predict the electrical conductivity of magnesium fluoride in each of the following states:

State	Electrical conductivity (Y / N)
Solid	
Liquid	
Aqueous	

[3 marks]

Question 3d

Explain why it is easier to use aqueous MgF_2 than molten MgF_2 in a laboratory setting.

[2 marks]

Question 4a

This question is about crystalline solids.

Iodine and diamond are crystalline solids at room temperature.

Explain why diamond has a much higher melting temperature than iodine.

[5 marks]

Question 4b

Graphite is also a crystalline solid at room temperature.

Unlike diamond, graphite conducts electricity.

Describe the key feature of the bonding of the carbon atoms in graphite that results in it being an electrical conductor.

[2 marks]

Question 4c

State the type of structure that is likely to be present in the following:

- i)
Crystalline solid that conducts electricity, is malleable and has a melting point of 900 °C.
- ii)
Crystalline solid that does not conduct electricity and has a melting point of 92 °C.
- iii)
Crystalline solid that does not conduct electricity and has a melting point of more than 3100 °C.
- iv)
Crystalline solid that does not conduct electricity and has a melting point of 790 °C. When molten it becomes a conductor.

[1]

[1]

[1]

[1]

[4 marks]

Question 5a

Carbon and silicon are both in Group 14.

Diamond and graphite are two forms of the element carbon.

Explain why diamond is a hard substance but graphite is soft.

[4 marks]

Question 5b

Two of the oxides of silicon and carbon are carbon dioxide, CO_2 , and silicon(IV) oxide.

Predict **two** differences in the physical properties of these two oxides.

[2 marks]

Question 5c

Silicon(IV) oxide is not soluble in water but magnesium oxide is soluble in water.

Explain why magnesium oxide is soluble in water.

[2 marks]

Model Answers: Easy

1a

a)

i) The arrangement and movement of particles in a gas are:

- Particles are far apart; [1 mark]
- Are in a random arrangement; [1 mark]
- Can move freely in all directions; [1 mark]

ii) Gases exert a pressure due to:

- Collisions between gas particles / molecules and the walls of the container; [1 mark]

[Total: 4 marks]

- When describing particles in any of the states of matter you should refer to:
 - How far apart the particles are:
 - Solids - touching each other
 - Liquids - close together
 - Gases - far apart
 - How the particles are arranged:
 - Solids - regular arrangement
 - Liquids - mostly irregular arrangement
 - Gases - randomly arranged
 - How the particles move:
 - Solids - in a fixed position but can vibrate
 - Liquids - able to move past each other in all directions
 - Gases - freely and in all directions
- You must refer to collisions when explaining why gases exert a pressure

1b

b) The relationship the graph in Fig. 1.1 shows between the volume of gas and pressure at

constant temperature is:

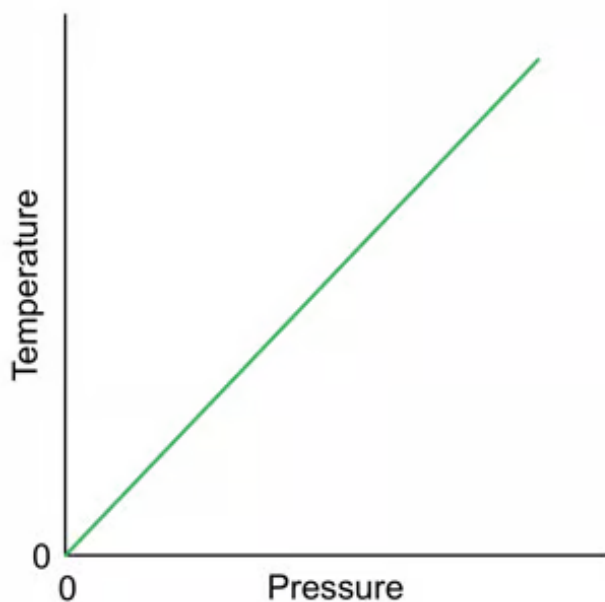
- Inversely proportional; [1 mark]

[Total: 1 mark]

- A straight line shows that the two variables are proportional
- This means that volume is proportional to $\frac{1}{\text{pressure}}$, so the volume is inversely proportional to pressure
- If a plot had been drawn of the volume of gas against pressure, a straight line with a negative gradient would show this relationship
- Decreasing the volume (at constant temperature) of the container causes the molecules to be squashed together
 - This results in more frequent collisions between the molecules and the container wall
 - So, the pressure of the gas increases

1c

c) The relationship between the temperature and the volume of the gas is:



- A straight line with a positive gradient; [1 mark]
- Line starting at the origin; [1 mark]

[Total: 2 marks]

- Increasing the temperature of a gas causes an increased collision frequency of the gas particles with the container wall so pressure increases
- Temperature is directly proportional to the pressure
- A directly proportional relationship is shown on a graph by a straight line that starts from the origin

2a

a) Helium comes very close to ideal gas behaviour because:

- It exists as single atoms

OR

It has a full outer shell so does not form bonds with other atoms; [1 mark]

- (It only has two electrons so) it has very weak / the smallest intermolecular forces of attraction / London dispersive forces / van der Waals' forces; [1 mark]

[Total: 2 marks]

- Hydrogen is another gas that comes close to ideal gas behaviour as it is even smaller than helium
- However, as it forms a hydrogen molecule with another hydrogen atom, the intermolecular forces are greater than those found between helium atoms

2b

b) The meaning of the terms p , V , and T , and the units are:

- p - the pressure of / exerted by the gas; [1 mark]
- Units of p - Pa **OR** Nm^{-2} ; [1 mark]
- V - the volume of the vessel (containing the gas)

OR

The volume occupied by the gas; [1 mark]

- Units of V - m^3 ; [1 mark]
- T - the (absolute) temperature of the gas; [1 mark]
- Units of T - K / kelvin / $273 + ^\circ\text{C}$; [1 mark]

[Total: 6 marks]

- It is very important to learn all of the units used for all of the terms in the ideal gas equation

- Equally important is knowing how to convert these units from more common units, e.g. cm^3 or dm^3 to m^3
- The most common mistakes students make with using the ideal gas equation are either forgetting to convert the units or converting incorrectly
- It is worth investing the time to learn how to do this
- Some common conversions are:
 - To convert from cm^3 to m^3 , divide by 10^6
 - Alternatively, you can multiply by 10^{-6} as this is the same calculation written in a different way
 - To convert from dm^3 to m^3 , divide by 1000
 - Alternatively, you can multiply by 10^{-3} as this is the same calculation written in a different way
 - To convert from kPa to Pa, multiply by 1000
 - To convert from $^{\circ}\text{C}$ to K, add 273

2c

c) The volume, in dm^3 , of one mole of helium gas at 10°C and 92.0 kPa is:

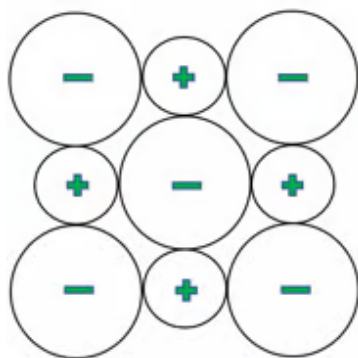
- $p = 92 \times 10^3\text{ Pa}$
AND
 $T = 283\text{ K}$; [1 mark]
- $V = \frac{1 \times 8.31 \times 283}{92 \times 10^3} = 0.02556\text{ (m}^3\text{)}$; [1 mark]
- $= 25.6\text{ (dm}^3\text{)}$; [1 mark]

[Total: 3 marks]

- One mark is available for correctly converting the units
- **Remember:** The answer will give a volume in m^3 but the question requires it to be given in dm^3
 - To convert from m^3 to dm^3 , multiply by 1000
- Don't forget to quote your final answer to 3 significant figures as prompted in the question
 - You would not be awarded the final mark if you didn't do this

3a

a) The ions in the lattice are positioned as:



- Alternating positive and negative charges; [1 mark]
- Positive charges in smaller circles

AND

Negative charges in larger circles; [1 mark]

[Total: 2 marks]

- Like charges repel, so ions with opposite charges are arranged in a lattice next to each other to minimise repulsions
- The question states that the sodium ions are the smaller ones, so you don't need to remember that, just read carefully!
- You would be awarded the mark if you put Na^+ and Cl^- instead of + and –

3b

b) The strength of the ionic bond in sodium chloride is due to:

- The (strong) electrostatic force / attraction; [1 mark]
- Between Na^+ and Cl^- ions / sodium ions and chloride ions / positive and negative ions; [1 mark]

[Total: 2 marks]

- This is a definition of ionic bonding that you need to know, using this specific language
- The example in the question is sodium chloride so you should talk about the sodium ion and the chloride ion
 - Mark schemes will often allow a more general statement about positive and negative ions, as in this mark scheme
 - However, this is not always the case so it is best to be specific

3c

c) The electrical conductivity of magnesium fluoride is:

State	Electrical conductivity (Y/N)
Solid	N; [1 mark]
Liquid	Y; [1 mark]
Aqueous	Y; [1 mark]

[Total: 3 marks]

- Ionic compounds will not conduct when solid as the ions cannot move
- Ionic compounds can conduct electricity when they are aqueous or liquid (molten) as the ions are able to move freely
- These rules apply to all ionic compounds so are easy to learn

3d

d) Aqueous magnesium fluoride is easier to use as:

- (MgF_2) has a high melting point; [1 mark]
- As there are (strong) electrostatic forces / attractions between oppositely charged ions in all directions; [1 mark]

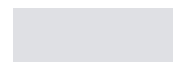
[Total: 2 marks]

- Ionic substances typically have high melting and boiling points
 - This is because the forces of electrostatic attractions across the 3D lattice are strong in every direction
 - Therefore, lots of energy is required to break the ionic bonds and melt / boil the compound
- Many ionic compounds are highly soluble at room temperature, so this is a much easier option to get a substance that will conduct electricity

4a

a) Diamond has a much higher melting temperature than iodine because:

- Iodine is (simple) molecular; [1 mark]
- Diamond is a giant molecular /covalent / lattice structure (with 4 covalent bonds per carbon atom); [1 mark]



- These rules apply to all ionic compounds so are easy to learn

3d

d) Aqueous magnesium fluoride is easier to use as:

- (MgF_2) has a high melting point; [1 mark]
- As there are (strong) electrostatic forces / attractions between oppositely charged ions in all directions; [1 mark]

[Total: 2 marks]

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4a

a) Diamond has a much higher melting temperature than iodine because:

- Iodine is (simple) molecular; [1 mark]
- Diamond is a giant molecular / covalent / lattice structure (with 4 covalent bonds per carbon atom); [1 mark]
- Iodine molecules are held together by weak London (dispersion) forces / dispersion forces / instantaneous induced dipole-dipole attractions; [1 mark]
- Carbon atoms in diamond are held together by (strong) covalent bonds; [1 mark]
- Strong covalent bonds require more energy to break than intermolecular forces / intermolecular forces require less energy to break than strong covalent bonds; [1 mark]

[Total: 5 marks]

- When explaining the properties of substances, always state the type of structure that the substance is
- Refer to the bonding / forces of attraction within the substance and how strong or weak these are

- Explain how these give rise to the particular property asked about
- If you are asked to explain the difference in property, as in this question, then make sure you use comparative language, i.e. the strong covalent bonds in diamond require more energy to break than the London dispersion forces in iodine
- Make sure you use the correct terminology, for example, any reference to diamond as a molecule will not gain a mark and neither would 'strong intermolecular forces'
- You would be allowed to use the term 'intermolecular forces' instead of London forces

4b

b) Describing the key feature of the bonding of the carbon atoms in graphite that results in it being an electrical conductor:

- One electron is free to move / delocalised (within the layer to carry the current); [1 mark]
- Each carbon is (covalently) bonded to three other carbons

OR

The carbon atoms are arranged in layers which allow the flow of electricity through them; [1 mark]

[Total: 2 marks]

- Be careful with terminology, just stating 'free electrons' will not gain the first mark, it is best to refer to the free electron in graphite as delocalised
- You could state that carbon uses three (outer shell) electrons in bonding
- You could gain both marks from a diagram that showed its structure

4c

c)

i) A crystalline solid that conducts electricity, is malleable and has a melting point of 900 °C is:

- Giant metallic (lattice); [1 mark]

ii) A crystalline solid that does not conduct electricity and has a melting point of 92 °C is:

- Simple molecular / covalent (lattice); [1 mark]

iii) A crystalline solid that does not conduct electricity and has a melting point of more than 3100 °C is:

- Giant covalent (lattice) / macromolecular; [1 mark]

iv) A crystalline solid that does not conduct electricity and has a melting point of 790 °C and when molten it becomes a conductor is:

- Giant ionic (lattice); [1 mark]

[Total: 4 marks]

- Being able to compare the properties of giant ionic, simple molecular, giant covalent and giant metallic lattices is key in this topic
- Electrical conductivity is a property present in molten ionic compounds, solid/molten metals and some giant covalent substances (graphite and graphene)
- As seen earlier in this question, a simple molecular structure that is a solid and melts at a relatively low temperature could be iodine for example

5a

a) Diamond is a hard substance but graphite is soft because:

- Diamond has a giant molecular / covalent lattice / macromolecular

OR

All the atoms are held in position / in tetrahedral structure / bonded to four other carbon atoms; [1 mark]

- It is difficult to break all of the strong covalent bonds; [1 mark]
- Graphite has layers of carbon atoms; [1 mark]
- The layers can slide over each other; [1 mark]

[Total: 4 marks]

- Comparing the properties of diamond and graphite is a common question
- **Remember:** In diamond, each carbon is bonded to four other carbon atoms, whereas in graphite, each carbon is bonded to only three, resulting in this difference in properties

5b

b) Two differences in the physical properties of these two oxides are:

Any **two** of the following:

- SiO_2 has a higher melting point / boiling point (than CO_2); [1 mark]
- SiO_2 is a solid (at room temperature and pressure)

AND

CO_2 is a gas (at room temperature and pressure); [1 mark]

- (When both are solids) then SiO_2 is harder; [1 mark]
- SiO_2 has a higher density (than CO_2); [1 mark]
- SiO_2 is insoluble

AND

CO_2 is soluble; [1 mark]

[Total: 2 marks]

- Carbon dioxide has a simple molecular structure whereas silicon(IV) oxide has a giant covalent lattice
- The differences in properties are caused by this difference in their structure
- You should be able to explain describe and predict the effect of different types of structure and bonding on the physical properties of substances, including melting point, boiling point, electrical conductivity and solubility

5c

c) Magnesium oxide is soluble in water because:

- Magnesium oxide / it has a giant ionic lattice; [1 mark]
- The ions on the surface are attracted to (polar) water molecules / form ion-dipole bonds; [1 mark]

[Total: 2 marks]

- When explaining the properties of a substance, always start by stating the type of structure it has, as this is usually worth a mark
- The ion-dipole attractions between the polar water molecules and the ions on the surface of the ionic lattice overcome the electrostatic attraction between the ions in the lattice

- This causes the ions to go into the solution

